

Reduced coordinates for the Collatz map: exact per-step laws, anchor dynamics, and the limits of counting arguments for cycles

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Abstract

We study the Collatz dynamics in a reduced coordinate system that compresses each deterministic valuation run into a single block, producing a self-map F on states (ω, d) — an odd core prime to 3 and a depth. The reduction is faithful: the Collatz conjecture is equivalent to every F -orbit reaching $(1, 1)$, with nontrivial cycles in bijection. In these coordinates the per-step arithmetic is exactly solvable. A single 2-adic quantity, the *anchor* $M(\omega) = -2 \log \omega / \log 9 \in \mathbb{Z}_2$, governs the step: the exit valuation obeys the global law $s = 2 + v_2(d - M(\omega))$ on the lifting classes and is constant on the others; the depth evolution closes exactly on every residue class; and the anchor increment along one step obeys an exact law modulo any power of 2, computable from graded residues of the state. A finite window of digits consequently decides each step in an error-free trichotomy, while a digit-budget argument shows no bounded window can decide infinite horizons — localizing the full difficulty of the problem in the digit supply of the anchors. On the cycle side, a one-line elimination identity yields short rederivations of the classical exclusions for cycles of one, two, and three blocks. Our main new theorem is a sharp dichotomy for counting arguments: a trim uniform in the number of blocks p exists, giving effective finiteness at every period, but its constant necessarily degrades like $(\log_2 3)^{-p}$ — an explicit family of near-counterexamples (*staircases*: geometric climbs closed by a single crash, precisely divergent-orbit profiles bent into loops) shows size-counting cannot do better, so uniform cycle exclusion requires arithmetic (divisibility) input, not sharper counting. Finally we state the equidistribution hypothesis implicit in the classical heuristics as a precise conjecture about an exactly computable product law, prove its consequences conditionally, and report a calibration campaign whose four apparent anomalies all dissolved under controls — one of them via an exact routing lemma that the biased estimator had been reflecting.

1 Introduction

Let $T(x) = (3x + 1)/2^{v_2(3x+1)}$ denote the odd-to-odd Collatz map on positive odd integers. The Collatz conjecture asserts that every T -orbit reaches 1. This paper develops a coordinate system in which the local arithmetic of T becomes exactly solvable, and uses it for three purposes: to prove exact per-step laws for the induced dynamics; to test how far counting arguments can

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go toward cycle exclusion (answer: exponentially far in the number of blocks, and provably no farther); and to state precisely the statistical hypothesis that the classical heuristics leave implicit.

The idea is to stop watching individual steps. Writing $x+1 = 2^m u$ with u odd, the next $m-1$ odd steps of T are deterministic — each converts one factor of 2 in $x+1$ into a factor of 3 — and terminate in a single cascade. Compressing this *block* and splitting $u = 3^a \omega$ with $3 \nmid \omega$ yields a state (ω, d) with $d = m + a$, and the block-to-block dynamics induce a self-map F on such states (Section 2). The compression loses nothing the conjecture depends on (Theorem 2.3).

The payoff is Section 3: the entire one-step arithmetic of F is governed by the 2-adic logarithm $M(\omega) = -2 \log \omega / \log 9 \in \mathbb{Z}_2$, which we call the *anchor*. The exit valuation, the 3-absorption, the next depth, and the low-order digits of the next anchor all obey exact laws in terms of the current anchor displacement $d - M(\omega)$. A finite window of state digits then decides the next step in a trichotomy that never errs (Theorem 3.5), and an elementary accounting argument (Proposition 3.6) shows each decision *consumes* digits irreversibly, so that bounded information cannot decide unbounded horizons. In these coordinates, the difficulty of the Collatz problem is not diffuse: it is exactly the question of where the digits of the anchors come from.

Section 4 turns to cycles, where finite data can in principle beat unbounded depth. A one-line elimination identity (Proposition 4.1) converts a hypothetical p -block cycle into p divisibility conditions on $q = 2^K - 3^n$, and short arguments rederive the classical exclusions of 1-, 2-, and 3-block cycles — results due in essence to Steiner [2] and Simons–de Weger [4], with the current record $m \leq 91$ by Hercher [5]; we claim no priority, only brevity. The main new result is Theorem 4.5 together with Theorem 4.6: a trim uniform in p exists, giving effective finiteness of the candidate set at every period, but its exponential degradation in p is *intrinsic* — the staircase family satisfies every size condition with room to spare and is killed only by divisibility. Since the staircase is precisely a divergent-orbit profile closed into a loop, the cycle half and the divergence half of the problem’s residual difficulty are, in a concrete sense, the same object.

Section 5 formalizes the statistical half. The folklore “fair-coin” heuristics (going back to Terras [1]) become, in reduced coordinates, an exact statement: orbit windows should equidistribute, *while values are large*, under an explicitly computable product law π_k whose consequences (the frequency ledger, the 3-gain rate $\frac{1}{3}$, the drift) are unconditional theorems *about* π_k . We report a calibration campaign with an unusual score: four apparent violations, four dissolutions under successively sharper controls, ending with bulk uniformity unqualified at all tested depths — and with an exact lemma (deterministic routing, Lemma 5.2) explaining how the last false anomaly aimed itself at a single cell.

Related work and provenance

Block-like decompositions and stochastic models are classical (Terras [1]; Everett; see Lagarias [6] for the literature), and the 2-adic conjugacy of the Collatz map to the shift underlies all Haar-genericity heuristics. Two-adic and Diophantine methods dominate the cycle literature [2, 4, 5, 3]; effective linear forms in logarithms [7, 8, 9] are the engine, and the verified range of the conjecture currently stands at 2^{71} [10]. Contemporary with this work, a documented human–LLM exploration [11] independently develops a burst–gap decomposition and an orbit equidistribution conjecture; it contains no anchor machinery, exact global laws, or cycle apparatus, and we became aware of it only during the final literature review. The division of labor in the present work is stated precisely, because it matters for both credit and blame. The reduced

coordinate system itself — the odd core (“seed”), the block cascade, the states (ω, d) , and the driving question of Section 3 — is the human author’s invention, originating in a hand-drawn note of 13 March 2023 that predates the collaboration. The analytical superstructure erected on it — the anchor formulation, the cycle apparatus, the equidistribution formalization, and the proofs — was developed by the AI collaborator under the author’s direction across three years of drafts, drawing on the classical literature’s concepts; any overlap with contemporary work lies in that layer, not in the coordinate system. Appendix A documents the process.

2 The reduced system

Definition 2.1. A reduced state is a pair (ω, d) with ω odd, $3 \nmid \omega$, $d \geq 1$. Set

$$A = 3^d \omega - 1, \quad s = v_2(A), \quad x_{\text{exit}} = A/2^s, \quad C = A + 2^s,$$

$$\sigma = v_2(C), \quad a_+ = v_3(C), \quad m_+ = \sigma - s, \quad \omega_+ = \frac{C}{2^\sigma 3^{a_+}}, \quad d_+ = m_+ + a_+,$$

and define the reduced map $F(\omega, d) = (\omega_+, d_+)$. The projection R sends an odd x with $x + 1 = 2^m 3^a \omega$ ($3 \nmid \omega$) to $R(x) = (\omega, m + a)$.

Every odd x determines a block of the T -dynamics: with $x + 1 = 2^m u$, the odd values rise deterministically for $m - 1$ steps (each step converting one factor of 2 of $x + 1$ into a factor of 3), then exit by the cascade dividing by 2^s . The block exit is $x_{\text{exit}}(R(x))$, and the projected next state is $F(R(x))$, independently of which representative of the state the orbit entered through. Full proofs of well-definedness are given in the repository (Appendix A); the statements we need are:

Proposition 2.2 (block structure). *The BlockEntry representatives of a state (ω, d) are exactly $(u, m) = (3^a \omega, d - a)$, $0 \leq a \leq d - 1$; the block from any representative runs deterministically to the exit $x_{\text{exit}}(\omega, d)$; and the sequence of reduced states visited by any T -orbit is the F -orbit of its projection.*

Theorem 2.3 (convergence translation). *$(1, 1)$ is a fixed point of F whose R -fiber is $\{1\}$. The Collatz conjecture holds if and only if every F -orbit from a valid state reaches $(1, 1)$, and R induces a bijection between nontrivial T -cycles and F -cycles other than $\{(1, 1)\}$.*

Proof sketch. The fiber statement is Proposition 2.2 with $x = 2^d \omega - 1$ representatives; both directions of the equivalence follow by tiling T -orbits with blocks, and the cycle bijection by periodicity of the deterministic entry–exit chain. (Checked exhaustively for the fiber over $x < 3 \cdot 10^6$ and along 300 random orbits.) $\square$

3 Anchor dynamics: the exact per-step laws

For $u \equiv 1 \pmod{8}$ the equation $9^t = u^{-1}$ has a unique solution $t \in \mathbb{Z}_2$ (as 9 topologically generates $1 + 8\mathbb{Z}_2$); write $N(u)$ for it, so $N(u) = -\log u / \log 9$ in the 2-adic logarithm. N is an isomorphism $(1 + 8\mathbb{Z}_2, \times) \rightarrow (\mathbb{Z}_2, +)$, and $v_2(9^t - 1) = 3 + v_2(t)$ for $t \neq 0$ (the *isometry*).

Definition 3.1 (anchor). *For every odd ω set $M(\omega) = N(\omega^2) = -2 \log \omega / \log 9 \in \mathbb{Z}_2$, well defined since $\omega^2 \equiv 1 \pmod{8}$. $M(\omega)$ is even iff $\omega \equiv \pm 1 \pmod{8}$.*

Theorem 3.2 (global valuation law). *Call (ω, d) lifting if $\omega \equiv 1 \pmod{8}$, d even, or $\omega \equiv 3 \pmod{8}$, d odd. On the lifting classes,*

$$s = v_2(3^d \omega - 1) = 2 + v_2(d - M(\omega)),$$

while on the six non-lifting residue-parity classes $s \in \{1, 2\}$ is constant, determined by $(\omega \bmod 8, d \bmod 2)$. Moreover $a_+ = 0$ iff s is odd (so the next state gains a factor of 3 exactly when s is even), and $s \leq C(\omega)(\log d)^2$ unconditionally by p -adic Baker theory [7, 8].

Theorem 3.3 (depth closure). *On every residue class the next depth closes exactly: $d_+ = m_+ + a_+$ where $a_+ = v_3(C)$ obeys an exact 3-adic absorption law and m_+ obeys the entry-depth law: on the lifting classes, with displacement ε ($= n - N(\omega)$ resp. $n - N(3\omega)$ for $d = 2n$ resp. $2n + 1$) and shift constant $\delta_s = N(1 - 2^s)$ (of exact valuation $s - 3$),*

$$m_+ = 3 - s + v_2(\varepsilon + \delta_s),$$

and off the lifting classes $m_+ = 1 + v_2(d - M(\omega))$ (the $s = 1$ classes) or $m_+ = v_2((d - 1) - M(\omega))$ (the $s = 2$ classes). The forced carry $v_2(\delta_s) = s - 3 = v_2(\varepsilon)$ reproves $m_+ \geq 1$ structurally.

Method of proof for 3.2–3.3. All statements reduce, through the isometry, to a single *target-shift lemma*: for $c \in 1 + 8\mathbb{Z}_2$, $v_2(3^{2n}\omega - c) = 3 + v_2(n - N(\omega) + N(c))$. The valuation law is the case $c = 1$; the entry-depth law is $c = 1 - 2^s$; the odd lifting component reduces to the even one via the companion parameter 3ω ; the off-lifting cases follow by a squaring trick ($v_2(X + 1) = v_2(X^2 - 1) - v_2(X - 1)$ with $X^2 \in 1 + 8\mathbb{Z}_2$). Each law was verified exhaustively (e.g. all 62,937 states $\omega < 3000$, $d < 64$ for the entry-depth law; 8,000 random states for the unified law) with zero discrepancies. $\square$

Theorem 3.4 (low-order anchor-increment law). *Let $\Delta M = M(\omega_+) - M(\omega)$. For every $k \geq 1$, $\Delta M \bmod 2^k$ is determined by, and explicitly computable from,*

$$\omega \bmod 2^{\sigma+k+2}, \quad d \bmod 2^{\sigma+k}, \quad a_+ \bmod 2^k,$$

and the stratum data (s, σ) are themselves determined by these residues. The moduli are fixed in advance on each stratum (s, m_+) .

Theorem 3.5 (one-step trichotomy). *From the depth- k window of Theorem 3.4 alone, one can determine: the exact next depth d_+ and next residue-parity class; the exact value $s_+ \in \{1, 2\}$ if the next class is non-lifting; and, if lifting, the truncation $\varepsilon_+ \bmod 2^k$ of the next displacement — yielding s_+ exactly when nonzero, and the correct report “ $s_+ \geq k + 2$: deep cascade ahead” when zero. The window never outputs a wrong value; the undecided rate under the product law is $\approx 2^{-(k+1)}$. (Verified along real orbits: 21,296 steps, 21,000 decided with zero errors, 296 flagged with zero bound violations.)*

Proposition 3.6 (digit budget). *Deciding one step at depth k consumes the state’s 2-adic data to depth $\sigma + k + 2$ (mean $\sigma \approx 4.0$ along orbits), and no bounded window regenerates it: a window of W digits supports $O(W/(k+6))$ decided steps. Bounded-window determinism along an infinite orbit is impossible in principle.*

Proposition 3.6 is the paper’s organizing negative result: it splits the residual difficulty into a statistical question for typical orbits (Section 5) and a rigidity question for cycles (Section 4), and nothing else.

4 Cycles: rederivations, a uniform trim, and its sharpness

Fix a hypothetical p -block cycle of F with entry depths $m_t \geq 1$, exit valuations $s_t \geq 1$ ($t \in \mathbb{Z}/p\mathbb{Z}$), and set $n = \sum m_t$, $K = \sum s_t + n$, $q = 2^K - 3^n$.

Proposition 4.1 (elimination identity). $q > 0$, and for every rotation r ,

$$\omega_r 3^{a_{r-1}} q = R_r := \sum_{t=0}^{p-1} 3^{M_t} 2^{S_t} (2^{s_t} - 1),$$

with M_t, S_t the rotation's partial m - and σ -sums. Hence $q \mid R_r$ and $q \leq \min_r R_r$. For the trivial cycle, $R = 4^p - 3^p = q$ identically.

Corollary 4.2 (size balance). $2^K = 3^n \prod_t (1 + \varepsilon_t)$ with $\varepsilon_t = (2^{s_t} - 1)/(3^{d_t} \omega_t)$; if every cycle element exceeds X then $0 < K \log 2 - n \log 3 < p/X$. This is the classical Baker-side engine [3], obtained in four lines.

Lemma 4.3 (ceiling). If $K > \lceil n \log_2 3 \rceil$ then some block exit is $< p/\ln 2$. Consequently $K = \lceil n \log_2 3 \rceil$ for every nontrivial cycle with $p \leq \ln 2 \cdot X_0$, where X_0 is the verified range [10] — unconditionally for $p \leq 5$ by inspection of the odd values ≤ 7 .

Theorem 4.4 (periods 1–3; rederivations). F has no nontrivial cycles of 1, 2, or 3 blocks. For $p = 1$ the fixed-point equation $\omega 3^a (2^{s+m} - 3^m) = 2^s - 1$ admits a window argument leaving at most one candidate s per m (the window's endpoint ratio is < 2); for $p = 2, 3$, budget-trim lemmas force $0 < K \log 2 - n \log 3 < 2^{c-n/5}$ resp. $2^{3-0.115n}$, exact searches to $n = 20,000$ find no solutions (the $p=3$ search collapses $\sim 10^{12}$ raw cells to 51 size-passers, none divisible), and any effective irrationality measure for $\log_2 3$ [9] excludes the tail. These results are contained in [2, 4, 5]; the derivations, not the theorems, are the contribution.

Theorem 4.5 (uniform trim). Every nontrivial p -block cycle satisfies

$$\gamma + \log_2 p > \frac{(\log_2 3 - 1)n}{(\log_2 3)^p - 1}, \quad \gamma := K - \log_2 q.$$

Consequently (with any effective measure for $\log_2 3$) the candidate set at every period is finite and explicitly bounded: $n \leq n_0(p) = O(p(\log_2 3)^p)$.

Proof sketch. For each rotation, $q \leq R_r < p 2^{\max_t e_t}$ with the exact exponent identity $e_t - K = (\log_2 3 - 1)M_t - m_r - \sum_{j>t} s_j$; the resulting per-block conditions $m_r < \gamma' + \Phi_r$ close into a cyclic max-plus recursion on arc weights $w(A) = (\log_2 3 - 1)m(A) - s(A)$, whose total is negative since $2^K > 3^n$; from the forced reset, partial sums grow at most geometrically with ratio $\log_2 3$. $\square$

Theorem 4.6 (sharpness: the staircase). The exponential degradation in Theorem 4.5 is intrinsic to size-counting. For configurations with block depths growing geometrically at ratio $\approx \log_2 3$ and unit exit valuations, closed by a single block of unit depth and maximal exit valuation, every rotation's size condition $q \leq R_r$ holds with $\gamma = O(\log p)$: e.g. at $p = 7$, $n = 94$, the staircase $m = (4, 7, 9, 15, 23, 35, 1)$ passes all seven size tests where any polynomial-in- p trim would forbid it (84 further examples at $p = 6$ alone). All such configurations fail the divisibility conditions $q \mid R_r$. Hence no counting argument uniform in p can improve the exponential constant, and uniform cycle exclusion requires the divisibility system — equivalently, rigidity of the closed anchor walk $\sum_t \Delta M_t = 0$.

Remark 4.7. The staircase profile — geometric depth growth with shallow exits — is exactly the growth regime of the size ledger for *divergent* orbits. Size analysis cannot forbid it as a cycle for the same reason drift analysis cannot forbid divergence. The two halves of the residual difficulty meet in one configuration.

5 The equidistribution hypothesis, made exact

Let π_k denote the product law on depth- k windows: ω -residues Haar-uniform among valid ones, depths following the stationary law of the exact window chain (an 8-state class skeleton with computable entries; e.g. from class $(\omega \equiv 1 \pmod{8})$, d odd) the next depth is exactly 1, and from $(\omega \equiv 5 \pmod{8})$, d odd), $P(d_+ \text{ even}) = \frac{2}{3}$ exactly). Under π_k , unconditionally: $P(s = j) = 2^{-j}$, the 3-gain rate is $\sum_{j \text{ even}} 2^{-j} = \frac{1}{3}$, and the classical drift follows — the folklore “ledger” is a theorem *about* π_k .

Hypothesis 5.1 (AEH, bulk form). *For every k and X , along the F -orbit of almost every state (natural density), the empirical distribution of depth- k windows over visits with $x_{\text{exit}} > X$ converges to π_k .*

AEH implies the ledger with error $O(2^{-k})$ via Theorem 3.5, the exact $\frac{1}{3}$ rate, and almost-everywhere contraction; it does not exclude individual staircase tails (Remark 4.7), and by Proposition 3.6 it cannot be reached by finite-window computation. Its content is a question about digits of 2-adic logarithms, older and broader than the Collatz problem.

Calibration. An eight-round campaign (per-orbit statistics, $\sim 1,600$ – $2,600$ independent orbits per cell) produced four apparent violations, each dissolved by a sharper control: an $s=3$ excess (small-start artifact), pair correlations (same), digit biases (contamination by the *bottom regime* — the fixed drainage basin of small integers, where deviations reach $z = 41$ but reflect particular numbers, not a measure), and finally a 5σ digit bias in a single cell surviving a 2^{40} cut. The last dissolved under fixed-horizon unweighted sampling (0.2503 ± 0.0015 against null 0.25, 84,739 visits) and was explained exactly:

Lemma 5.2 (deterministic routing). *For $\omega \equiv 1 \pmod{8}$, $d = 1$: the residue $\omega \bmod 32$ determines the next class exactly — $1 \mapsto \text{itself}$ (immediate self-return), $9 \mapsto (7, 1)$, $17 \mapsto (5, 1)$, $25 \mapsto (3, 1)$, the lifting class.*

Ratio estimators over correlated visit sequences inherit this routing (residue 25 leads to deep descent, hence fewer qualifying visits, hence inflated per-orbit weight), manufacturing a bias aimed at one residue of the unique class with a self-loop channel. Bulk uniformity stands unqualified at all tested depths.

6 Discussion

In reduced coordinates the Collatz problem factors cleanly. The per-step arithmetic is solved: one 2-adic logarithm governs everything local, and finite windows decide steps without error. What remains is a single question wearing two costumes: for typical orbits, whether the anchors’ digits equidistribute (Hypothesis 5.1 — statistics); for cycles, whether closed anchor walks are rigid (Theorem 4.6 — arithmetic). The staircase shows the costumes cover the same body. We

see three directions with defined success criteria: rigidity statements for closed anchor walks beyond the size level (the only route past Theorem 4.6); transfer-operator analysis of the exact window chain; and, on the statistical side, any progress on digits of 2-adic logarithms of integers, a problem this paper inherits rather than creates.

A Methodology and records

This work was produced by a human–AI collaboration over three years (2023–2026), from a hand-drawn origin note (13 March 2023) through seventy-nine drafts to a research wiki whose pages carry status headers, verification records, and refuted claims with their refutation data. The human author contributed the coordinate system, the research direction, and the standards of evidence; the AI contributed the mathematical development within it — formulations, proofs, computations, and the connections to the classical literature — and this asymmetry is disclosed rather than averaged. Every computational claim above cites a runnable script; every “proved” result was re-verified with independently written code before being so labeled; the calibration campaign’s four dissolved anomalies, and two corrected search-filter errors on the cycle side, are documented rather than smoothed over. The complete record (wiki, code, draft history, origin note) is available from the author. Precedent for documenting LLM-assisted mathematical exploration: [11].

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